

Mukuru

Data Analytics Internship, Technical Assessment

Submitted by **Phemelo Sebopelo** in response to the Mukuru Data Analytics Internship assessment. This report covers the full analytical solution for Questions 1, 2 and 3, using SQL, Python, Excel and Power BI.

Candidate	Phemelo Sebopelo
Contact	063 313 6167 phemelosebopelo14@gmail.com
Role Applied	Data Analytics Intern, Mukuru
Date	March 2026
Tools	SQL (PostgreSQL) Python (Pandas) Excel Power BI

1. Executive Summary

Mukuru's remittance business was on a strong upward trajectory between April 2012 and March 2013. Across 13 months of transaction data, paid order volumes grew in every year-on-year comparison period, active customers expanded by 53%, and roughly 6 in every 10 customers came back to send money the following month. These numbers tell a story of a product that people trust and keep coming back to.

Five things that stand out from the data:

- Paid orders grew over 1,000% year-on-year in April 2012, reflecting the scale of expansion happening at that time.
- March 2013 was the strongest month on record with 38,336 paid orders, a +40.4% MoM spike worth investigating for replication.
- Active customers grew from 15,458 to 23,648 across the period, steady, consistent growth every single month.
- The Repeat Sender Rate averaged 62.7%, the majority of customers are habitual senders, not one-time users.
- Zimbabwe and Malawi account for 99% of payout volume, a corridor concentration that represents both strength and risk.

2. Analytical Approach

2.1 Tools & Process

The analysis followed a structured process: explore the data, validate quality, write SQL logic, verify calculations in Python, build Excel workings, and visualise in Power BI.

- SQL (PostgreSQL) for core filtering, aggregation and metric calculation logic
- Python (Pandas) for independent verification of all results and dataset exploration
- Excel for structured workings with live formulas showing calculation logic
- Power BI for the interactive dashboard with slicers for business users

2.2 Filtering Rules Applied

All metrics exclude cancelled orders and orders with unpaid sentinel dates, as defined in the assessment glossary.

- **Paid Order:**

OrderPaid contains a valid date, excluding sentinel values 1900-01-01, 1970-01-01 and 0.

- **Cancelled Order:**

Excluded where OrderCancelled contains a valid non-sentinel date.

- **Reporting Window:**

Last 13 months, April 2012 to April 2013.

- **Partial Month (April 2013):**

Data runs to 14 April only. MoM and YoY use MTD vs MTD comparisons per the glossary.

2.3 Metric Definitions

MoM (%)	$(\text{Current Month} - \text{Prior Month}) / \text{Prior Month} \times 100$
YoY (%)	$(\text{Current Month} - \text{Same Month Prior Year}) / \text{Same Month Prior Year} \times 100$. April 2013 uses MTD vs MTD.
Active Customers	3-month rolling count of distinct SenderKeys who paid for at least one order
Monthly Repeat Senders	Customers who transacted in both the current month AND the preceding month
Repeat Sender Rate (%)	$\text{Monthly Repeat Senders} / \text{Transacting Customers in preceding month} \times 100$

3. Question 1, Paid Orders Over Time

The Commercial team wants to understand paid order volumes and how they are changing month-on-month and year-on-year. Results cover the last 13 months with filters for PayIn Country, PayOut Country and Date.

3.1 Power BI Dashboard, Q1

The dashboard below shows monthly paid orders from April 2012 to April 2013, with MoM and YoY % changes in the table visual. Slicers allow the Commercial team to filter by PayIn Country, PayOut Country and Month.



3.2 Results Table

Detailed monthly breakdown of paid orders, MoM change and YoY change:

Month	Paid Orders	MoM Change (%)	Prior Year Orders	YoY Change (%)
Apr 2012	19,272	N/A	1,741	+1,006.9%
May 2012	24,787	+28.6%	4,000	+519.7%
Jun 2012	22,169	-10.6%	4,864	+355.8%
Jul 2012	25,164	+13.5%	6,308	+298.9%
Aug 2012	24,170	-4.0%	8,288	+191.6%
Sep 2012	22,952	-5.0%	8,512	+169.6%
Oct 2012	25,638	+11.7%	10,115	+153.5%
Nov 2012	25,808	+0.7%	12,040	+114.4%
Dec 2012	30,978	+20.0%	14,808	+109.2%
Jan 2013	31,327	+1.1%	15,633	+100.4%
Feb 2013	27,305	-12.8%	15,469	+76.5%
Mar 2013	38,336	+40.4%	18,523	+107.0%
Apr 2013*	12,158	-68.3%*	7,900*	+53.9%

* April 2013 is a partial month, data runs to 14 April only. The MoM decline of -68.3% reflects the incomplete month, not a real business drop. YoY uses April 2012 orders up to day 14 (7,900 orders) for a fair like-for-like comparison.

3.3 Insights

The business was in hyper-growth mode

Every month in this period delivered positive year-on-year growth. Over 1,000% YoY in April 2012, over 500% in May. That's a business that was barely operational twelve months prior and is now scaling fast. By late 2012 the YoY rates settle into the 76–114% range, still exceptional, reflecting a more established but still rapidly growing base.

March 2013 stands alone

38,336 paid orders in a single month, up 40.4% from February. Nothing else in the dataset comes close. The question worth asking is: what happened in March 2013? A new corridor launch, a marketing push, end-of-financial-year remittance behaviour? Identifying and replicating this driver would be high-value work for the Commercial team.

Mid-year softness is a recurring pattern

June, August and September show consistent MoM declines. This mid-year dip signals a seasonal opportunity, targeted promotions during Southern African winter months could help drive volume when remittance behaviour naturally softens.

December is predictably strong

December 2012 saw a clean +20.0% MoM surge. The festive season remittance spike is predictable and plannable. The Commercial team should have promotional campaigns and liquidity buffers ready for it every year.

Zimbabwe corridor concentration is a strategic risk

Zimbabwe accounts for 72% of all paid orders. Malawi adds 27%. That's 99% of payout volume in two countries. Any regulatory disruption affecting the Zimbabwe corridor would hit total volumes hard. Corridor diversification is both a growth opportunity and a risk management imperative.

4. Question 2, Active Customers & Monthly Repeat Senders

The Customer Journey team wants to understand the size of the active and repeat sender base over time. Results cover the last 13 months with filters for PayIn Country and PayOut Country.

4.1 Power BI Dashboard, Q2

The dashboard below shows the 3-month rolling active customer trend, monthly repeat sender volumes, and the repeat sender rate over time. Slicers allow filtering by PayIn Country and PayOut Country.



4.2 Active Customers Results

Month	Active Customers (3-Month Rolling)	Month-on-Month Change
Apr 2012	15,458	Baseline
May 2012	16,851	+9.0%
Jun 2012	17,606	+4.5%
Jul 2012	18,680	+6.1%
Aug 2012	19,102	+2.3%
Sep 2012	19,393	+1.5%
Oct 2012	19,665	+1.4%
Nov 2012	20,114	+2.3%

Month	Active Customers (3-Month Rolling)	Month-on-Month Change
Dec 2012	21,272	+5.8%
Jan 2013	22,305	+4.9%
Feb 2013	22,814	+2.3%
Mar 2013	23,648	+3.7%
Apr 2013*	21,791	-7.8%*

4.3 Monthly Repeat Senders Results

Month	Transacting Customers	Prev Month Customers	Repeat Senders	Repeat Sender Rate (%)
Apr 2012	9,420	8,712	5,072	58.2%
May 2012	11,244	9,420	6,134	65.1%
Jun 2012	10,949	11,244	6,801	60.5%
Jul 2012	12,050	10,949	7,056	64.4%
Aug 2012	11,951	12,050	7,384	61.3%
Sep 2012	11,526	11,951	7,150	59.8%
Oct 2012	12,489	11,526	7,381	64.0%
Nov 2012	12,746	12,489	7,740	62.0%
Dec 2012	14,299	12,746	8,506	66.7%
Jan 2013	14,667	14,299	9,239	64.6%
Feb 2013	13,612	14,667	8,812	60.1%
Mar 2013	15,721	13,612	9,095	66.8%
Apr 2013*	7,954	15,721	5,677	36.1%*

* April 2013 figures reflect a partial month (14 days only). The Repeat Sender Rate of 36.1% is artificially suppressed, the denominator is a full March month while the numerator covers only 14 days of April activity.

4.4 Insights

Active customer growth never reversed

From April 2012 to March 2013, eleven consecutive months, active customers grew every single month. A 53% increase from 15,458 to 23,648 in under a year tells you Mukuru was not just acquiring customers but retaining enough of them to grow the 3-month rolling base consistently.

Six in ten customers come back every month

The Repeat Sender Rate averaged 62.7% across the 12 complete months. More than half of every month's active base are returning senders who used the service last month and came back again. This level of habitual behaviour is the foundation of a sustainable remittance business.

March 2013 was exceptional across every customer metric

March 2013 simultaneously recorded the highest active customers (23,648), highest transacting customers (15,721), highest repeat sender count (9,095) and second-highest repeat sender rate (66.8%). When volume, base size and loyalty all peak together, it's a signal worth understanding deeply.

December brings back lapsed customers

December 2012 recorded the highest Repeat Sender Rate in the dataset at 66.7%. The festive season doesn't just attract high volumes, it re-engages customers who may have skipped a month. December is not just a volume opportunity, it's a re-activation opportunity.

5. Question 3, Data Quality Validation

If given access to similar data in a real work environment, here is how I would approach validating its quality, grounded in what I actually observed working through this specific dataset.

5.1 Start with the data dictionary

Before touching the data, read the dictionary. Every field, every relationship, every glossary definition. This is how I identified the sentinel date convention in this dataset, 1900-01-01, 1970-01-01 and 0 are used to represent null states in timestamp fields. Without understanding that, my paid order counts would have been wrong by 118,220 records.

5.2 Confirm all expected datasets are present

Before any analysis starts, verify that every table exists, contains data, and covers the expected date range. During this assessment, the Orders and Users files were initially missing, in a real environment this is equivalent to a pipeline not loading or a table not refreshing.

5.3 Check for structural anomalies

Look for things that shouldn't be there. This dataset contained four completely null columns, Unnamed: 13 through Unnamed: 16, across all 568,638 rows. These serve no analytical purpose but could cause confusion if not identified early. Also confirm data types match dictionary definitions.

5.4 Validate referential integrity

Every foreign key in Orders should have a matching record in the relevant dimension table. Every PayInCountryKey should exist in Country, every SenderKey should match a UserKey in Users. Orphaned keys mean orders exist for customers or countries with no reference record, a data pipeline failure that corrupts dimension-level analysis.

5.5 Apply business logic checks

- OrderPaid should never be earlier than OrderCreated, you cannot pay before an order exists
- OrderCompleted should always be after OrderPaid, funds cannot be collected before they are sent
- An order should not be both paid and cancelled, mutually exclusive states
- PayInValue should be positive and non-zero for any paid order

5.6 Assess volume plausibility

Botswana appears as a PayIn country with only 138 orders versus 568,500 for South Africa, 0.02% of total volume. Before reporting any Botswana insight I would verify whether this reflects a new market, an incomplete extract, or a data mapping issue. Any unexpected spike or collapse in monthly volume warrants investigation before being reported as a real business event.

5.7 Document and communicate findings

Data quality findings affect the reliability of every insight that follows. I document every anomaly, its scale, and its potential impact, then communicate to the data engineering team and business stakeholders before finalising any analysis. Where anomalies cannot be resolved, I apply consistent handling rules and make them visible in every report and dashboard.

"Good analysis built on unvalidated data is still bad analysis. The quality of the insight is only as strong as the quality of the data behind it."

6. Conclusion

The data tells a clear story: between April 2012 and March 2013, Mukuru's remittance business was growing fast, retaining customers well, and building a loyal base of repeat senders across its Southern African corridors. The Zimbabwe corridor dominates current volume, a core strength to protect and a concentration risk to manage. March 2013 stands out as the peak of performance across every metric, understanding what drove it could shape Mukuru's commercial strategy going forward.

This analysis was built on clean, validated data with every calculation defined explicitly against the assessment glossary. The SQL scripts, Excel workings and Power BI dashboard submitted alongside this report provide the full analytical foundation, with interactive slicers enabling the Commercial and Customer Journey teams to explore results by PayIn Country, PayOut Country and date as needed.

Phemelo Sebopelo

063 313 6167 | phemelosebopelo14@gmail.com

github.com/Phemelocodemode | linkedin.com/in/phemelo-sebopelo